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Brief Article

Profiling the Fungal Microbiome after Fecal Microbiota Transplantation for Graft-versus-Host Disease: Insights from a Phase 1 Interventional Study

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A B S T R A C T

Disruption of the intestinal bacterial microbiota is frequently observed in the context of allogeneic hematopoietic cell transplantation (HCT) and is particularly pronounced in patients who develop graft-versus-host disease (GVHD). Donor fecal microbiota transplantation (FMT) restores gut microbial diversity and reduces GVHD in HCT recipients. The composition of the intestinal fungal community in patients with GVHD, and whether fungal taxa are transferred during FMT are currently unknown. We performed a secondary analysis of our clinical trial of FMT in patients with steroid-refractory GVHD with a focus on the mycobiota. We characterized the fecal mycobiota of 17 patients and healthy FMT donors using internal transcribed spacer amplicon sequencing. The donor who provided the majority of FMT material in our study represents an n-of-one study of the intestinal flora over time. In this donor, mycobiota composition fluctuated over time while the bacterial microbiota remained stable over 16 months. Fungal DNA was detected more frequently in baseline stool samples from patients with steroid-refractory GVHD than in patients with steroid-dependent GVHD. We could detect fungal taxa in the majority of samples but did not see evidence of mycobiota transfer from donor to recipient. Our study demonstrates the feasibility of profiling the mycobiota alongside the more traditional bacterial microbiota, establishes the methodology, and provides a first insight into the mycobiota composition of patients with GVHD.

INTRODUCTION

Graft-versus-host disease (GVHD) is an aberrant anti-host immune response that causes high morbidity and mortality among patients undergoing allogeneic hematopoietic cell transplantation (HCT) [1,2]. Disruption of the intestinal bacterial microbiota is frequently observed in the context of allogeneic HCT and is particularly pronounced in patients that develop GVHD [3]. Fecal microbiota transplantation (FMT) restores gut microbial diversity and reduces GVHD in allogeneic HCT recipients [4]. The mechanism of action of FMT is unknown, even in *Clostridioides difficile* infection (CDI), where there is good evidence of efficacy [5].

Although live bacteria are thought to be the biologically active component, other components of the fecal microbiome, such as fungal communities, may contribute to efficacy [6]. In patients with ulcerative colitis as well as in patients with recurrent CDI, the abundance of intestinal fungi (ie, the fungal

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microbiome or "mycobiome"), particularly *Candida* species, is associated with FMT efficacy [7,8]. The composition of the intestinal fungal community in patients with GVHD, and whether fungal taxa are transferred during FMT, are currently unknown.

Allogeneic HCT recipients frequently develop fungal infections, caused predominantly by *Candida* and *Aspergillus* species [9,10]. Our group recently showed that the mycobiome composition of allogeneic HCT recipients has high day-to-day variability, and that fungal dysbiosis, defined by the presence of culturable fungi or domination of the fungal mycobiome with a single fungal species, is associated with worse outcome after allogeneic HCT [11]. Previous work suggests that colonization of *Candida* species within days after admission for allogeneic HCT is associated with a higher frequency of acute GVHD [12], and that prophylactic use of fluconazole is associated with a lower incidence of severe gut GVHD [13]. Mycobiota composition can be determined in stool samples using high-resolution sequencing of the fungal internal transcribed spacer (ITS) 1, a fungal DNA barcode.

We sought to examine the mycobiota in a single-center cohort of FMT donors and recipients using the sample processing and analysis pipeline that we previously established [14], with the goal of understanding the dynamics and clinical importance of the mycobiota in patients with GVHD in relation to FMT.

METHODS

Study Design and Patient Population

Stool samples for the present study were collected in a previous phase 1 trial studying the effect of donor FMT on GVHD symptoms (ISRCTN 14530574). Patient selection criteria and the FMT procedure have been described previously [4]. Patients received antifungal prophylaxis according to institutional protocols (Table 1).

Analysis of Fecal Samples

16S bacterial amplicon sequencing and fungal amplicon sequencing were performed as described previously [11]. For fungal sequencing, stool samples were processed with lyticase prior to bead beating and phenol-chloroform. The ITS1 region was amplified with Phusion polymerase, and sequencing was performed on the Illumina MiSeq platform (Illumina, San Diego, CA). The DADA2 R package was used to infer the exact amplicon sequencing variants. Taxonomic annotation was performed using a National Center for Biotechnology Information BLAST-based algorithm [15] with the full UNITE+INSD database [14,16] and compiled with the *phyloseq* package. Mycobiota analyses were performed using R version 3.6.2 (R Foundation for Statistical Computing, Vienna, Austria).

Statistics

Calculated values are denoted as mean ± standard deviation. Statistical significance was determined with the Mann-Whitney *U*-test using GraphPad Prism software or R.

RESULTS

Seventeen patients with biopsy-proven, steroid-refractory (n = 8) or steroid-dependent (n = 9) acute GVHD each received a single FMT via nasoduodenal infusion from an unrelated, healthy donor (Table 1). Steroid-refractory GVHD was defined as intestinal GVHD that did not respond to prednisolone therapy (2 mg/kg/day for acute GVHD), showed no improvement in overall grade of GVHD within 7 days or showed progression of at least 1 grade within the first 72 hours after starting high-dose prednisolone treatment. Steroid-dependence was defined as repeated recurrence of intestinal GVHD during steroid tapering. Infection prophylaxis was continued during the trial, including antifungal medications in 11 of 17 subjects (Supplementary Table S1).

FMT alleviated symptoms of both steroid-refractory (patient [P] 11, P16, and P17) and steroid-dependent GVHD (P01, P02, P06, P07, P09, P12, and P13) . Fungal DNA was undetectable in 6 of 9 patients with steroid-dependent GVHD, whereas it was detected in all 8 baseline samples from steroid-refractory patients (Figure 1A). In addition to the more frequent presence of fungal DNA, the steroid-refractory patients had more severe intestinal GVHD compared with the steroid-dependent study patients (stage 3.0 ± 0.9 versus 1.3 ± 0.7; *P* = 0.001), higher prednisolone dosage at the day of FMT (1.4 ± 0.8 mg/kg versus .5 ± 0.3 mg/kg; *P* = 0.02, not including P03, P09, and P11, who were not receiving prednisolone at the time of FMT), and all received total parenteral nutrition, compared with only 1 steroid-dependent patient (P07). In 4 of the steroid-refractory patients, the fungal microbiome was dominated by a single genus or species (P04, P10, P11, and P17), including 2 cases of *Candida* species domination (P10 and P11) (Figure 1A).

Sixteen subjects received FMT of stool collected on the day of FMT, from the same healthy donor (donor 2). The bacterial microbiota of this donor remained highly stable over time (Figure 1B), with the majority from the phyla Actinobacteria and Firmicutes (and orders Ruminococcaceae and Lachnospiraceae). In contrast, the fungal microbiome appeared to be

Table 1
Patient Characteristics

Characteristic	Value
Total patients, n	17
Age, yr, median (range)	57 (20-72)
Sex, female/male, n	10/7
FMT indication, steroid refractory/dependent, n	8/9
Time between HCT and FMT, mo, median (range)	7 (1.6-49.9)
Diagnosis, n	
Acute myelogenous leukemia	9
Myelodysplastic syndrome	3
Hodgkin lymphoma	1
Non-Hodgkin lymphoma	1
Myeloproliferative disorder, myelofibrosis	3
Graft source, n	
Peripheral blood stem cells	16
Cord blood	1
Donor type, n	
Matched unrelated	15
Matched related	2
Conditioning type, n	
Reduced intensity	13
Myeloablative	4
GI GVHD stage at start of FMT, n	7
II	3
III	4
IV	3
GI GVHD treatment other than systemic steroids at start of FMT, n	
Local therapy	9
Systemic therapy	13
GVHD involvement of other organs at time of FMT, n	
Skin	4
Liver	2
Antifungal treatment at start of FMT, n	
Fluconazole	6
Posaconazole	3
Voriconazole	2

GVHD staging was performed according to internationally accepted criteria [25,26].
GI indicates gastrointestinal.

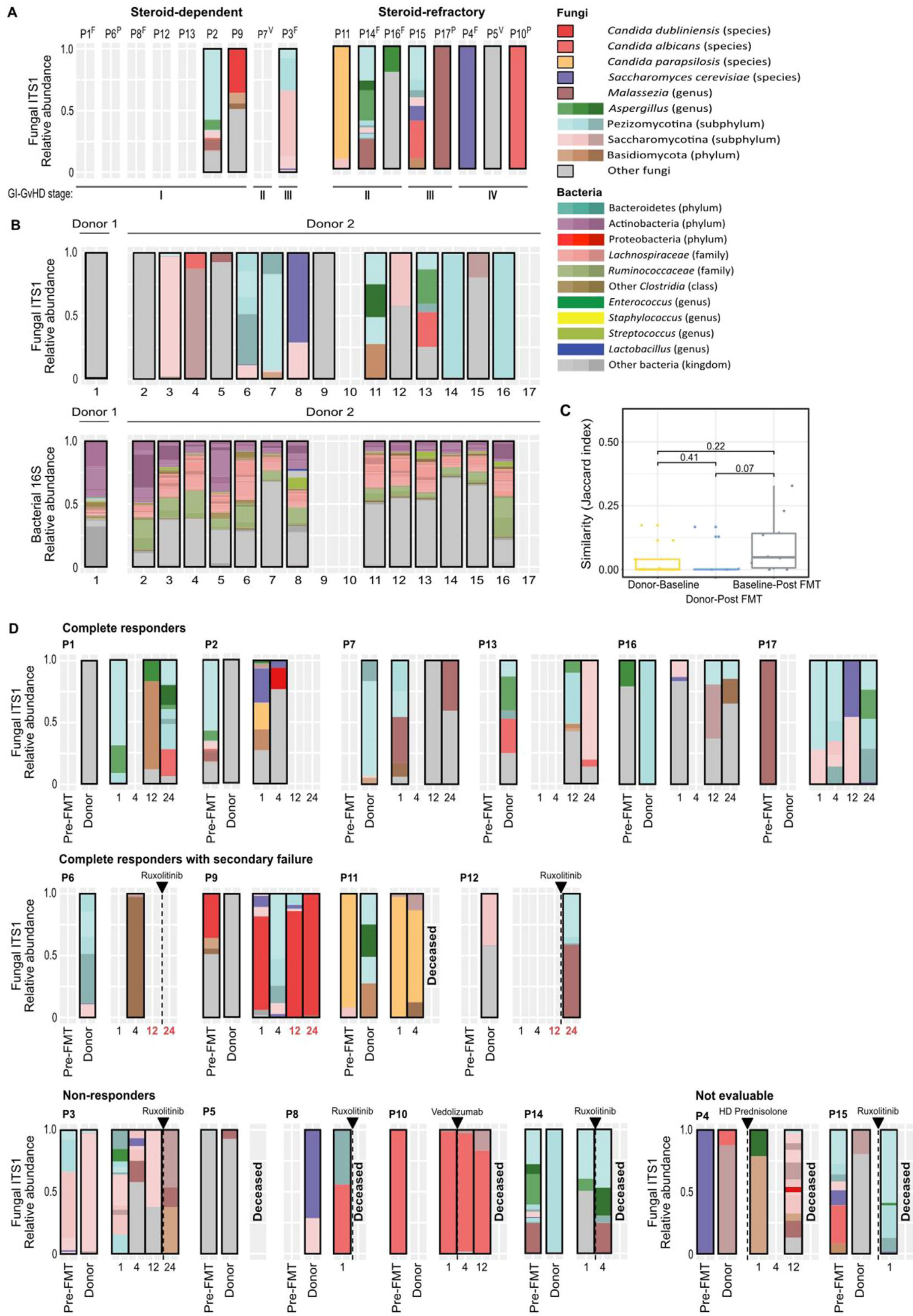

Figure 1. Mycobiota and bacterial microbiota dynamics in FMT-treated GVHD patients and healthy FMT donors. (A) Fecal mycobiota composition in allogeneic HCT recipients with steroid-dependent or steroid-refractory patients pre-FMT, aligned by GVHD severity. Study participants are indicated by numbers above each bar graph; superscript letters specify the antifungal prophylaxis received by the respective patient: F, fluconazole; P, posaconazole, V, voriconazole. Fungal taxa are colored according to the legend; for samples that failed to produce an ITS-amplicon, only the background (light- gray rectangles with white gridlines) is shown without a colored bar. (B) Relative abundance of fungal species (top row) and bacterial species (bottom row) in healthy donors for FMT. Samples are chronologically aligned, spanning a period of 16 months for donor 2. (C) Boxplots indicating similarity (Jaccard index) between the patients' baseline mycobiota composition and that of their FMT donor (n = 9), patients' composition 1 week after FMT and that of their FMT donor (n = 11), and the patients' composition at baseline and at 1 week after FMT (n = 9). (D) Composition of the mycobiota in all stool samples collected during the 6-month follow-up after FMT treatment from complete responders (top row), complete responders with secondary failure (middle row), and nonresponders and unevaluable patients who received extra GVHD medication prior to response evaluation (bottom row). The numbers below each bar graph indicate the time point of sampling (weeks after FMT); red numbers represent the time of GVHD recurrence in complete responders with secondary failure. The black dashed line and arrow indicate the timing of new GVHD medication prescribed in cases of FMT failure. HD,

highly variable over time, as described previously [17,18]. *Candida* and/or *Aspergillus* species were detected in 3 donated samples that were administered to patients P04, P11, and P13. Neither of these species could be detected in the first available patient's stool sample at 1 week after donor FMT. In contrast to our previous bacterial analysis in which we observed that the proportion of donor bacterial species detected in FMT recipient samples increased in association with clinical response [4], we could not confirm consistent engraftment of donor-derived fungal DNA in these samples. Although there was some similarity between the fungal microbiome of patients with GVHD before and 1 week after FMT, the respective donor sample had low similarity (measured using the Jaccard index) with the sample at 1 week after FMT (Figure 1C). Only 2 amplicon sequencing variants were detected in 2 donor-recipient pairs, but the sequences were not well described in the UNITE database. We could not detect the transmission of a potential pathogenic fungus in any instance; however, owing to the sampling schedule, we were unable to assess any transient fungal engraftment in the early days following FMT. Transfer of donor-derived fungal taxa has been reported in a previous case study describing a single transplantation recipient undergoing repeat FMT for GVHD. In that patient, serial samples were collected between 5 and 120 days after each procedure, with fungal taxa examined using metagenomic sequencing. Engraftment appeared transient, with high proportions of donor-derived taxa early after FMT, and a decrease to near-undetectable during the follow-up period after 3 of the 4 FMTs performed [19].

Finally, we investigated the fungal microbial community composition of patients grouped by the response of GVHD to FMT treatment (Figure 1D); however, with a cohort of this size, we were unable to observe any differences in community composition between the responders and nonresponders.

DISCUSSION

Fungal DNA was detected more frequently in baseline stool samples from patients with steroid-refractory GVHD than in patients with steroid-dependent GVHD. Assuming that selective technical failure is unlikely, this suggests that fungal DNA is either absent in patients with steroid-dependent GVHD or that the fungal biomass in these patients is below the limit of detection. The long-term reliance of patients with steroid-dependent GVHD on immunosuppressants generally extends their exposure to antifungal medication, which may eradicate (a proportion of) their intestinal mycobiota. Moreover, the patients with steroid-refractory GVHD received higher prednisolone dosages at baseline. The bacterial microbiome analysis in our previous study demonstrated that steroid-refractory patients exhibited lower bacterial diversity than steroid-dependent participants (Shannon index, 3.3 ± .6 versus 2.2 ± .9; P = .02). The sample size of this study was too small to correlate the presence or absence of specific fungal species to the administration of antifungal therapy or changes in bacterial diversity. In allogeneic HCT recipients, decreased bacterial load and low bacterial diversity were previously associated with high abundances of fecal *Candida* species [20]. In mice, it has been demonstrated that outgrowth of pathogenic *Candida* species can be facilitated by antibiotic eradication of bacterial populations [21,22]. In those studies, administration of prednisolone increased the fungal burden in the mouse gastrointestinal tract, lowered the IFN-γ and IL-17A cytokine concentrations, and induced extensive mucosal damage in the small intestine [21]. A recent study found that yeast colonized injured tissues of patients with Crohn's disease and prevented mucosal healing in experimental models [23]. Taken together, these findings suggest a possible mechanistic relationship between steroid administration, fungal burden in the gastrointestinal tract, and inflammation. In a recent study from our group, fungal dysbiosis was associated with decreased overall survival and higher transplantation-related mortality [11]. It remains to be determined whether a dysbiotic mycobiota in HCT recipients is also associated with other outcomes, such as relapse of primary disease or development of (chronic) GVHD.

A single healthy donor provided the majority of fecal transplants used in this study. This provided a unique opportunity to study the stability of the healthy bacterial and fungal microbiota over the 16 months during which this individual volunteer provided fecal samples. In contrast to the bacterial microbiota composition, which was highly stable, consistent with previous literature [24], the composition of the fecal mycobiota fluctuated over time. The variability in fungal community composition seen in both donor 2 and our treated patients requires further exploration and may reflect the susceptibility of fungi to environmental change, including those induced by diet and medication changes. We were unable to demonstrate fungal "engraftment" post-FMT and thus were unable to assess whether fungal transfer contributes to FMT efficacy. This will be an important issue to address in future FMT studies to optimize FMT donor selection.

Here we have demonstrated that fungal DNA can be measured in the majority of stool samples collected from patients with GVHD and was more frequently detected in subjects with severe, steroid-refractory GVHD. Although our study was underpowered to allow for any definitive conclusions, it would be of interest to analyze the impact of these findings on outcomes in larger cohorts. Our study highlights the promise of studying fungal communities in the intestinal tract in the setting of FMT and indeed in HCT recipients in general. Definitive studies of the fungal microbiome are needed and likely will contribute to the rational design of microbiota-targeted therapies for HCT recipients in the future.

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high dose. Mycobiota analyses were performed using R version 3.6.2 (R Foundation for Statistical Computing, Vienna, Austria). Calculated values are denoted as mean ± standard deviation. Statistical significance was determined with the Mann-Whitney U test using R or Prism (GraphPad Software, San Diego, CA).

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Authorship statement: Y.F.v.L. designed the study, collected clinical data, and wrote the manuscript; T.R. analyzed and interpreted the sequencing data and contributed to figure preparation; B.Z. performed research and contributed to critical data analysis and interpretation; G.K.A. performed the DNA extraction and prepared the amplicon libraries for sequencing; N.J.E. H. performed sample preparation; E.M. and E.N. enrolled study participants; B.B., J.U.P., M.R.M.v.d.B., and T.M.H. contributed to the conception and editing of the manuscript; and K.A.M. and M.D.H. designed the study, were major contributors to data interpretation and edited the manuscript. All authors had access to the study data and reviewed and approved the final manuscript. Y.F.v.L. and T.R. contributed equally as co-first authors. M.D.H. and K.A.M. contributed equally as co-last authors.

SUPPLEMENTARY MATERIALS

Supplementary material associated with this article can be found in the online version at doi:10.1016/j.jtct.2022.10.011.

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